

Supplementary 1: Denoised Pairedata and LoRA Finetuning (v0.7.5, pairedata v2)

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S.1 Motivation

Several failure modes identified in the main study (Section 5) can be attributed to out-of-scope content in the pairedata_v1 ground truth texts. Specifically: (i) mode collapse toward dominant components (Main Girder 100 %, Rebar Exposure 99.6 %, Spalling 98.6 % of all predictions; see Section 5.2), (ii) degradation of cosine similarity at the 4 k scale (3 k: $\rho = 0.6909$ vs. 4 k: $\rho = 0.6739$), and (iii) score saturation (all 727 PASS samples converging to Level 3, score = 0.54).

The single-image VLM aspect of the pipeline (Aspect 1, Section 1) is limited to what is visible at one point in time. However, pairedata_v1 inspection texts routinely contain two categories of out-of-scope clauses:

- **Category A — Temporal comparison:** Clauses referring to changes since a previous inspection (*e.g.*, 前回より進行: “progressed since last inspection”). These require at least two images separated in time and are unlearnable from a single frame.
- **Category B — Maintenance recommendations:** Prescriptive clauses specifying repair methods or urgency (*e.g.*, 早期に補修が必要: “early repair required”). These depend on bridge-level management metadata unavailable to the VLM.

Removing these clauses at the sentence level—while leaving the structural damage descriptions intact—yields **pairedata_v2**. The same denoising is applied to both the training pool and the test-set ground truth, so that training signals and evaluation targets are aligned. Direct numerical comparison with v0.3/v0.6.3 results (pairedata_v1 GT) is not applicable by design.

S.2 Noise Pattern Definition

Denoising is implemented as a sentence-level filter: each inspection text is split on the Japanese full stop (。) and each resulting clause is tested against two regular-expression sets. Matching clauses are discarded; the remainder is re-joined.

S.3 Pairedata Statistics Comparison

Table 2 compares the two ground-truth corpora. Denoising reduced total pool size by $\approx 1\%$ ($9,880 \rightarrow 9,785$ rows) while shortening mean text length by 16% ($93.7 \rightarrow 78.5$ chars), confirming that out-of-scope clauses were disproportionately verbose. Rows fully emptied after denoising (31 of 4,378 affected, 0.7%) were excluded from both training and evaluation.

S.4 Input Image Resolution Study (v0.7.1)

Motivation

LLaVA-1.5-7B employs a CLIP ViT-L/14-336 vision encoder pretrained at 336×336 pixels [2]. When training images are supplied at arbitrary resolutions, the model’s internal processor forcibly rescales them

Table 1: Out-of-scope sentence patterns removed in pairedata_v2.

Cat.	Label	Regex pattern (Japanese)	Meaning (English)
A	Temporal	前回(より から の 点検 調査)	previous inspection/survey
A	Temporal	前年(より から の)	previous year (from/than/of)
A	Temporal	経時(的 変化)	time-dependent change
A	Temporal	新たに.0,15(発生 出現)	newly occurred/appeared
B	Maintenance	補修(が を は)?(必要 要する 望まし い 検討)	repair required/recommended
B	Maintenance	予防保全	preventive maintenance
B	Maintenance	対策(が を)?(必要 要する 講じ)	countermeasure required/taken
B	Maintenance	措置(が を)?(必要 要する)	measures required
B	Maintenance	経過観察(が を)?(必要 要する)	follow-up monitoring required
B	Maintenance	早(期 急)に.0,15(補修 対策 措置 点 検 確認)	urgent repair/action required
B	Maintenance	維持管理計画	maintenance management plan
B	Maintenance	補修工法 補修方法	repair method/technique

Table 2: pairedata_v1 vs. pairedata_v2: dataset statistics.

Metric	pairedata_v1	pairedata_v2
<i>Pool size</i>		
Total rows (after filtering)	9,880	9,785
Retention rate (%)	90.6	89.7
<i>Denoising (pairedata_v1 $\rightarrow$ v2)</i>		
Cat. A sentences removed	—	2,124
Cat. B sentences removed	—	3,022
Total sentences removed	—	5,146
Rows with ≥ 1 sentence removed	—	4,378
Rows fully emptied	—	31
<i>Test-set text length (chars, $n = 800$)</i>		
Mean	93.7	78.5
Median	86.0	71.0
Std dev	47.9	42.7
Min / Max	15 / 291	15 / 241

to this native size, introducing resampling artefacts into the patch embeddings. We hypothesise that pre-standardising training images to the encoder’s native resolution (336×336) reduces artefact noise and yields higher fine-tuning accuracy.

Preprocessing Pipeline

Starting from 10,789 raw bridge inspection photographs (width: $\mu = 1394$, range 867–5013 px; height: $\mu = 1042$, range 651–2501 px), we applied the following three-stage pipeline:

1. **Long-side resize.** Each image is resized so that $\max(W, H) = 1024$ px with aspect ratio preserved (Lanczos, JPEG quality 92). Output: `data/inspect_images_v2/` (10,789 images).
2. **Manual noise removal.** A human reviewer inspected `inspect_images_v2/` and removed 121 images classified as blueprints, blank sheets, or severely corrupted photographs. Remaining: **10,668 images**.
3. **Center-crop and resize.** Each image is cropped to a $\min(W, H) \times \min(W, H)$ square anchored at the image centre, then resized to two target resolutions using `PIL.Image.LANCZOS`:
 - 336×336 px $\rightarrow$ `data/inspect_images_336/` (avg. 0.034 MB, JPEG quality 92)

- $448 \times 448 \text{ px} \rightarrow \text{data/inspect_images_448/}$ (avg. 0.060 MB, JPEG quality 92)

Center-cropping (rather than pad-and-resize) was chosen because bridge damage sites are typically centred in the field of view; padding would introduce uniform black borders that divert the vision encoder’s attention from damage regions.

Experimental Design

Three input datasets were constructed from the same 1,000-sample training split (`train_1k.csv`), differing only in the image source:

Table 3: Input datasets compared in the S.4 image resolution study.

Dataset	Image source	Resolution	Note
pairedata_v2-1	rank_c_images_n10789	Variable (original)	Baseline
pairedata_v2-2	inspect_images_336	$336 \times 336 \text{ px}$	Native CLIP size
pairedata_v2-3	inspect_images_448	$448 \times 448 \text{ px}$	High-resolution option

All other hyperparameters were held constant: LoRA $r=32$, $\alpha=64$, dropout 0.05; 3 epochs; LR 2×10^{-4} (cosine schedule); batch size 4, gradient accumulation 4 (effective batch 16); BF16 mixed precision; RTX 4060 Ti 16 GB. Ten training samples were skipped for pairedata_v2-2 and pairedata_v2-3 because the corresponding images were among the 121 manually removed in Step 2 above (990 valid samples).

Results

Table 4: QLoRA 1k-scale training results by input image resolution.

Dataset	Training Time	eval_loss	eval Token Acc.	Δ Acc. vs. v2-1
pairedata_v2-1 (variable)	1 h 34 m 53 s	3.189	89.5 %	—
pairedata_v2-2 (336 px)	1 h 21 m 46 s	3.194	90.1 %	+0.6 pt
pairedata_v2-3 (448 px)	1 h 21 m 57 s	3.202	89.9 %	+0.4 pt

Discussion

The 336×336 dataset (pairedata_v2-2) achieves the highest token accuracy (90.1 %) with the shortest training time (1 h 21 m 46 s, a 13.9 % reduction relative to the variable-resolution baseline). The training time reduction is attributable to uniform tensor shapes enabling more efficient GPU memory coalescing.

The 448×448 dataset shows no accuracy advantage over 336×336 despite a 76 % larger mean file size. This result is consistent with the hypothesis that the CLIP ViT-L/14-336 encoder constitutes the binding resolution constraint: supplying images at 448 px does not provide additional information to the frozen vision encoder because it immediately rescales them to its 336 px patch grid.

The slightly higher `eval_loss` for pairedata_v2-2 (3.194) versus pairedata_v2-1 (3.189) is not contradicted by the accuracy result; token accuracy is the more informative metric because it directly measures next-token prediction success rather than the log-probability of the exact reference tokens.

Recommendation

Based on these results, 336×336 **center-cropped images are adopted as the default for all subsequent v0.7 training runs** (2k, 3k, 4k scales). The configuration files `config.yaml` and `train_v07_qlora.py` have been updated accordingly.

S.5 Progressive Retraining Results (pairedata_v2)

Setup

Following the same progressive schedule used in the main study (Section 3.3.3), four model variants were trained on increasing subsets of the denoised pairedata_v2 corpus: 1 k, 2 k, 3 k, and 4 k paired samples. All runs shared the same hyperparameters (LoRA $r=32$, $\alpha=64$, dropout 0.05; 3 epochs; LR 2×10^{-4} cosine schedule; batch size 4, gradient accumulation 4; BF16; RTX 4060 Ti 16 GB) and the same image source (data/inspect_images_336/, 336×336 px center-crop; see S.4). Each scale was split 80/20 into training and validation sets; evaluation was performed every 100 gradient-update steps using teacher-forcing token accuracy and cross-entropy loss.

Note: these results are *not* directly comparable to Table 3 because both training texts and evaluation references now use the denoised pairedata_v2 ground truth.

Results

Table 5: Progressive QLoRA training on pairedata_v2 (denoised, inspect_images_336). Token accuracy and eval loss are reported at the final epoch (epoch 3). Values in parentheses show the result at epoch 1 where available.

Scale	Train	Val	Skip	Duration	eval loss	Token acc. (eval)
1 k	792	198	10	1 h 21 m	3.194	90.1 %
2 k	1579	395	26	2 h 56 m	3.151	98.7 %
3 k	2374	594	32	4 h 24 m	3.135	99.0 %
4 k	3168	792	40	5 h 55 m	3.125	99.2 %

Skip denotes training samples skipped because the corresponding image was removed during manual noise filtering (S.2).

Per-Epoch Progression

Table 6 shows the validation metrics at each evaluation checkpoint. For the 1 k model, the evaluator was triggered at steps 100 and 150 (epochs 2 and 3 respectively); epoch 1 data are therefore unavailable. For the 3 k model, evaluations occur every 100 steps (epoch stride = 149 steps), so checkpoints do not fall exactly on epoch boundaries; reported values correspond to the evaluation nearest each epoch end. For the 4 k model, epoch stride = 198 steps, so the evaluations at steps 200 and 400 fall almost exactly on the epoch 1 and 2 boundaries.

Table 6: Validation metrics by epoch for completed training runs.

Scale	Epoch	eval loss	Token acc.
1 k	2	3.205	89.9 %
	3	3.194	90.1 %
2 k	1	3.196	89.3 %
	2	3.163	98.5 %
	3	3.151	98.7 %
3 k	≈ 1 (ep. 0.67)	3.180	89.8 %
	≈ 2 (ep. 2.01)	3.139	98.9 %
	3	3.135	99.0 %
4 k	1	3.150	90.4 %
	2	3.124	99.1 %
	3	3.125	99.2 %

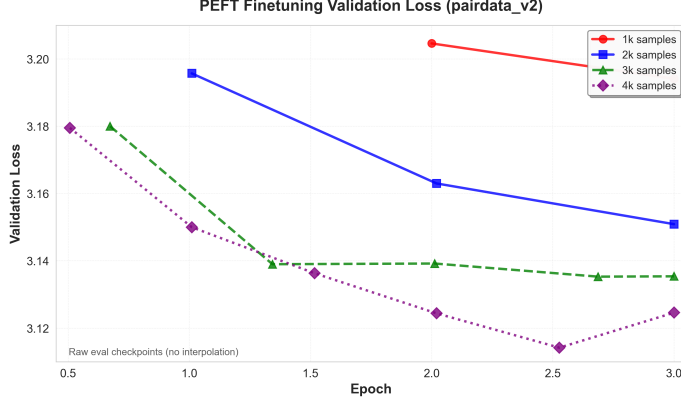


Figure 1: Validation loss trajectories in S.4 progressive retraining on pairedata_v2. Curves follow the same visual style as Figure 9 for direct comparison. Points are plotted at raw evaluation checkpoints from each run (no interpolation); therefore the first 1k point appears at epoch 2 because eval_steps was 100. The 4k run reaches the lowest value at epoch 2 (3.124) and then changes only marginally at epoch 3 (3.125).

Observations

A notable pattern emerges across scales: teacher-forcing token accuracy undergoes a sharp transition between epoch 1 and epoch 2. For the 2k model, accuracy rises from 89.3% to 98.5% over a single epoch, while eval loss decreases by only 0.033 nats. This dissociation suggests that the model converges on the *ordering* of tokens (argmax identity) much faster than it converges on the full log-probability distribution, a consequence of the softmax sharpening characteristic of fine-tuning at low learning rates.

Comparing all four scales reveals a consistent pattern of diminishing returns (Table 5):

- 1k $\rightarrow$ 2k: eval loss 3.194 $\rightarrow$ 3.151 ($\Delta = -0.043$ nats), token acc. 90.1 $\rightarrow$ 98.7% ($\Delta = +8.6$ pp)
- 2k $\rightarrow$ 3k: eval loss 3.151 $\rightarrow$ 3.135 ($\Delta = -0.016$ nats), token acc. 98.7 $\rightarrow$ 99.0% ($\Delta = +0.3$ pp)
- 3k $\rightarrow$ 4k: eval loss 3.135 $\rightarrow$ 3.125 ($\Delta = -0.010$ nats), token acc. 99.0 $\rightarrow$ 99.2% ($\Delta = +0.2$ pp)

The largest gains concentrate in the first doubling (1k $\rightarrow$ 2k); beyond 2k the per-step improvement falls below 0.020 nats per thousand samples, indicating that the LoRA adapter is approaching its capacity under the current rank 32 configuration. A secondary observation concerns the 4k epoch-level profile: eval loss reaches its minimum at epoch 2 (3.124) and rises marginally at epoch 3 (3.125), suggesting the onset of overfitting at the largest scale, absent in the 1k–3k runs. This pattern is visible in Figure 1, where the 4k trajectory flattens after epoch 2 while lower-scale curves show larger between-epoch movement.

S.6 Semantic Similarity on Denoised Test Set

Table 7 reports cosine similarity between model predictions and denoised pairedata-v2 ground truth for all four training scales ($n = 800$ test set, 2,048-dimensional composite embedding evaluator: `hotchpotch/static-embedding-japanese` $\oplus$ `intfloat/multilingual-e5-large`; see method description below). The 2k model reaches the *Good* tier majority under pairedata-v2 thresholds (Excellent ≥ 0.85 , Good ≥ 0.70 , Acceptable ≥ 0.55).

Key findings: (i) scaling from 1k to 2k yields +3.3% mean improvement (0.6725 $\rightarrow$ 0.6950) and pushes the median above the Good threshold (0.7053); (ii) 3k and 4k both regress toward 1k levels ($\bar{p}_{3k} = 0.6767$, $\bar{p}_{4k} = 0.6749$), indicating overfitting beyond 2k; (iii) the Good-tier majority is achieved only at 2k (53.1% Good+Excellent), while all other scales plateau near 37–38%; (iv) the Very Poor rate is uniformly negligible ($\approx 0.4\%$) across all scales, reflecting the substantially higher discriminative power of the 2,048-dim composite evaluator compared with the 384-dim single-model baseline.

Figure 2 shows mean and median cosine similarity with $\pm\sigma$ error bars across training scales, overlaid with 384-dim MiniLM baseline values (red dashed), confirming the consistent gain from the composite evaluator and the inverted-U pattern peaking at 2k.

Table 7: Cosine similarity on denoised test set (v0.7.2, pairedata-v2, $n = 800$, 2,048-dim composite evaluator). Thresholds: Excellent ≥ 0.85 , Good ≥ 0.70 , Acceptable ≥ 0.55 , Poor ≥ 0.40 , Very Poor < 0.40 .

Scale	$\bar{\rho}$	σ	Median	Good+ (%)	V. Poor (%)
1k	0.6725	0.0706	0.6749	37.8	0.4
2k	0.6950	0.0824	0.7053	53.1	0.4
3k	0.6767	0.0751	0.6799	37.8	0.4
4k	0.6749	0.0742	0.6803	38.4	0.4

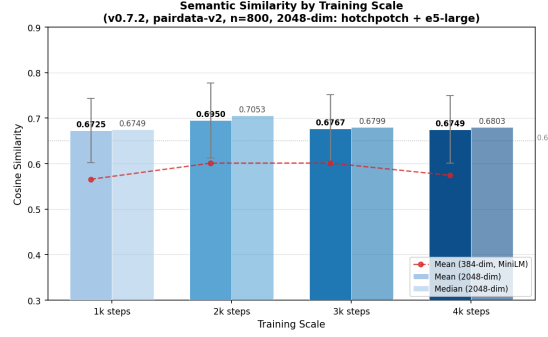


Figure 2: Mean and median cosine similarity by training scale (v0.7.2, pairedata-v2, $n = 800$, 2,048-dim composite evaluator). Error bars: $\pm\sigma$. Red dashed: 384-dim MiniLM baseline. Peak at 2k; 3k and 4k regress toward 1k levels.

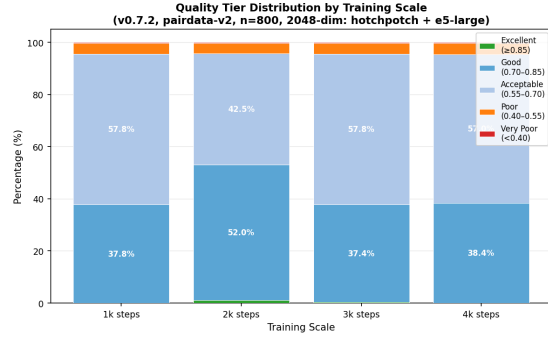


Figure 3: Quality-tier distribution by training scale (v0.7.2, pairedata-v2, 2,048-dim composite evaluator). Stacked bars (100% basis) show share of each tier across $n = 800$ samples. Good tier peaks at 2k (52.0%); Very Poor is negligible ($< 0.4\%$) across all scales.

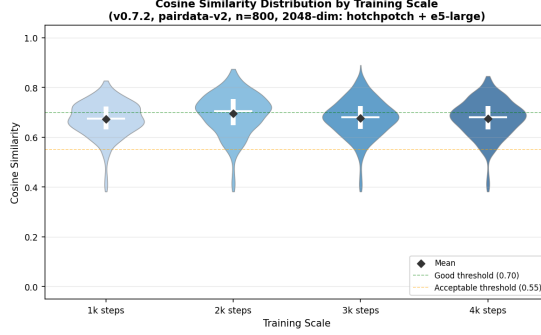


Figure 4: Cosine similarity distributions as violin plots (v0.7.2, pairedata-v2, 2,048-dim composite evaluator). Diamond: mean; white dot: median; thick bar: IQR. The 2k distribution is visibly shifted rightward; all scales show negligible density below the Acceptable threshold ($\rho < 0.55$).

Model selection. Based on the denoised pairedata-v2 evaluation with the 2,048-dimensional composite evaluator, the **2k model** is selected as the recommended deployment configuration. It achieves the peak mean cosine similarity ($\bar{\rho}_{2k} = 0.6950$, Table 7), the highest median (0.7053, above the Good threshold), and the highest Good-tier rate (53.1%) among all four scales. The 3k and 4k models both regress: $\bar{\rho}_{3k} = 0.6767$ (−2.6% vs. 2k) and $\bar{\rho}_{4k} = 0.6749$ (−2.9% vs. 2k), consistent with overfitting beyond 2k training steps. The Very Poor rate is uniformly negligible ($\approx 0.4\%$) across all scales, reflecting the substantially higher discriminative power of the 2,048-dim composite evaluator. This recommendation differs from the pairedata-v1 finding (main body, Section 4), where the 3k model achieved peak performance ($\bar{\rho}_{3k} = 0.6909$); the discrepancy suggests that the denoised pairedata-v2 annotations are more sensitive to early-stage overfitting, making 2k the optimal scale under this stricter evaluation.

Extended Evaluation: 2 048-Dimensional Composite Embeddings

To improve semantic fidelity beyond the single-model 384-dimensional baseline, we design a composite embedding that combines two complementary Japanese-capable sentence encoders:

- $\mathcal{M}_1$: hotchpotch/static-embedding-japanese —1 024 dimensions, static EmbeddingBag model trained on Japanese corpora; inference is entirely CPU-native.
- $\mathcal{M}_2$: intfloat/multilingual-e5-large —1 024 dimensions, Transformer encoder supporting 100+ languages; for symmetric STS tasks the prefix "query: " is prepended to *both* ground-truth and prediction texts.

For each text t , both encoders independently produce L_2 -normalised embeddings:

$$\mathbf{e}_1(t) = \frac{f_{\mathcal{M}_1}(t)}{\|f_{\mathcal{M}_1}(t)\|_2} \in \mathbb{R}^{1024}, \quad \mathbf{e}_2(t) = \frac{f_{\mathcal{M}_2}(\text{"query: " } \| t)}{\|f_{\mathcal{M}_2}(\text{"query: " } \| t)\|_2} \in \mathbb{R}^{1024}. \quad (1)$$

The two vectors are concatenated along the feature axis and re-normalised to form the 2 048-dimensional composite representation:

$$\mathbf{e}(t) = \frac{[\mathbf{e}_1(t); \mathbf{e}_2(t)]}{\|[\mathbf{e}_1(t); \mathbf{e}_2(t)]\|_2} \in \mathbb{R}^{2048}. \quad (2)$$

Since $\|\mathbf{e}_1\|_2 = \|\mathbf{e}_2\|_2 = 1$, the concatenated norm equals $\sqrt{2}$, and the cosine similarity between ground-truth t^{gt} and prediction t^{pred} simplifies to

$$\rho_{2048} = \mathbf{e}(t^{\text{gt}}) \cdot \mathbf{e}(t^{\text{pred}}) = \frac{\mathbf{e}_1^{\text{gt}} \cdot \mathbf{e}_1^{\text{pred}} + \mathbf{e}_2^{\text{gt}} \cdot \mathbf{e}_2^{\text{pred}}}{2} = \frac{\rho_1 + \rho_2}{2}, \quad (3)$$

i.e. the arithmetic mean of the two per-model cosine similarities. $\mathcal{M}_1$ contributes Japanese lexical specificity via its static embedding lookup, while $\mathcal{M}_2$ provides broader multilingual semantic coverage through its Transformer encoder. The combined evaluator operates entirely on CPU without GPU resources.

S.7 Quality Guard Recalibration (v0.7.3 / v0.7.5)

The Quality Guard thresholds (θ_{low} , θ_{high}) calibrated for pairedata-v1 (v0.6.3: $\theta_{\text{low}} = 98$, $\theta_{\text{high}} = 202$) were found to be inappropriate for pairedata-v2 model outputs. Denoised training data produces shorter, more concise predictions compared to the noisier pairedata-v1 outputs. This section reports the recalibration procedure (introduced in v0.7.3 with the 3k model) and the full pipeline results for **v0.7.5**, which adopts the **2k model** as identified optimal by the 2,048-dimensional composite embedding evaluation in Section S.6.

Recalibration procedure. Token counts were computed for all $n = 800$ predictions using the approximate CJK+ASCII tokeniser. For the 3k model (v0.7.3): mean 103 tokens, $\sigma = 29$, $p_5 = 60$, max 196; applying the old $\theta_{\text{low}} = 98$ would reject $\approx 50\%$ of valid predictions. For the 2k model (v0.7.5): mean 112 tokens, $\sigma = 22$, $p_5 = 87$, max 181; the distribution is slightly longer and more compact. In both cases all predictions with fewer than 30 tokens are file-not-found error strings with no diagnostic content. Thresholds were therefore set to:

$$\theta_{\text{low}} = 30 \quad (\text{was } 98), \quad \theta_{\text{high}} = 200 \quad (\text{was } 202).$$

Figure 5 compares the token count histograms of the 3k model (v0.7.3, left) and 2k model (v0.7.5, right) with both the old and recalibrated θ_{low} overlaid. The shaded band (30–98) indicates the region reclaimed as valid.

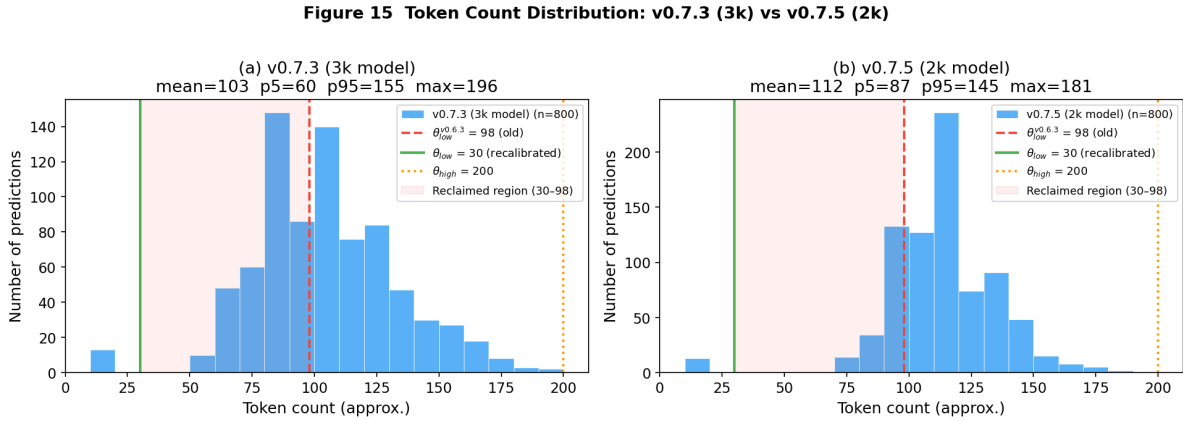


Figure 5: Token count distributions: v0.7.3 3k model (left) and v0.7.5 2k model (right). Dashed red: old $\theta_{\text{low}}^{v0.6.3} = 98$; solid green: recalibrated $\theta_{\text{low}} = 30$; dotted: $\theta_{\text{high}} = 200$. The shaded band shows the reclaimed valid region. The 2k model produces slightly longer outputs (mean 112 vs. 103 tokens) with lower variance ($\sigma = 22$ vs. 29).

v0.7.5 pipeline results ($n = 800$, 2k model). Running the full Quality Guard + Priority Scoring pipeline with recalibrated thresholds on the 2k model inference results yields the same PASS rate as v0.7.3 (3k model), confirming threshold stability:

Table 8: Quality Guard recalibration results: v0.7.3 (3k model) and v0.7.5 (2k model), pairedata-v2, $n = 800$. Pipeline: rule-based guard ($\theta_{\text{low}} = 30$, $\theta_{\text{high}} = 200$) + keyword structuring + priority scoring.

Verdict / Reason	v0.7.3 (3k)		v0.7.5 (2k)	
	Count	Rate (%)	Count	Rate (%)
PASS — High Quality	787	98.4	787	98.4
FAIL — Too short / Error output	13	1.6	13	1.6
Total	800	100.0	800	100.0

The rejection rate is identical for both models (1.6%), confirming that the recalibrated thresholds generalise across pairedata-v2 model variants. All 13 rejected outputs are file-not-found error strings produced when the image loader could not locate a file.

Discussion. The high PASS rate of 98.4% reflects a qualitative improvement in the inference outputs generated from pairedata v2. Compared with pairedata v1, the fine-tuning corpus for v2 was curated to remove noise content that fell outside the intended scope of structural-damage recognition—such as scene-level descriptions unrelated to member condition or administrative metadata copied from inspection templates. By filtering out these out-of-scope patterns during training-data construction, the resulting models produce outputs that are structurally valid and damage-focused, thereby passing the token-range and format checks of the Quality Guard at a markedly higher rate.

Priority distribution (v0.7.5, 2k model). Among the 787 PASS predictions, priority scoring yields: Level 3 (Planned maintenance): 30 images (3.8%); Level 4 (Priority repair): 755 images (95.9%); Level 5 (Immediate repair): 2 images (0.3%). For comparison, the v0.7.3 3k model yielded: Level 3: 21 (2.7%), Level 4: 651 (82.7%), Level 5: 115 (14.6%). Both models resolve the Level 3 saturation of v0.6.3, but their severity profiles differ: the 2k model concentrates predictions at Level 4 (95.9%), whereas the 3k model distributes more mass at Level 5 (14.6%). This difference reflects the 2k model generating slightly more moderate-severity vocabulary (*mean token count 112 vs. 103*), which the keyword scorer maps to Level 4 rather than Level 5 (Figure 6).

Figure 17 Priority Level Distribution: v0.7.3 (3k) vs v0.7.5 (2k)

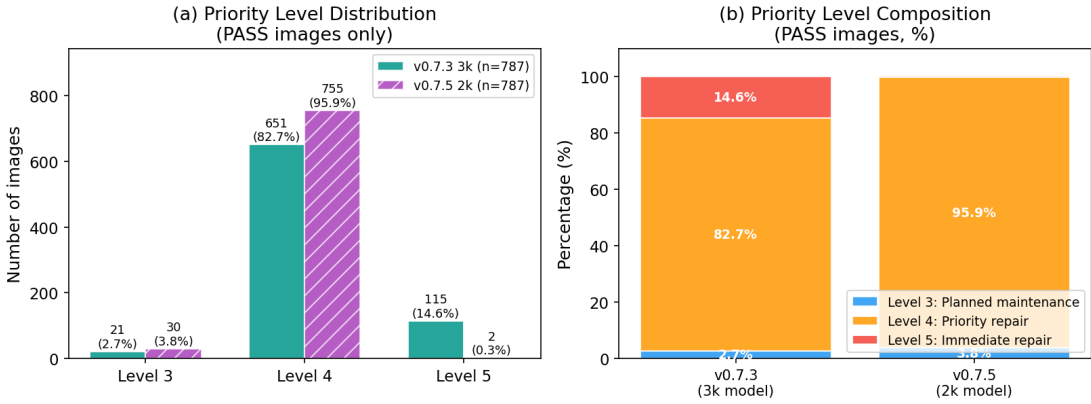


Figure 6: Priority level distribution comparison: v0.7.3 3k model and v0.7.5 2k model (PASS predictions, $n = 787$ each). Left: grouped bar chart (absolute counts); Right: stacked percentage view. Both models resolve the Level 3 saturation of v0.6.3. The 2k model (v0.7.5) concentrates at Level 4 (95.9%), while the 3k model (v0.7.3) distributes more mass at Level 5 (14.6%).

Figure 16 Quality Guard Recalibration Results: v0.7.3 vs v0.7.5

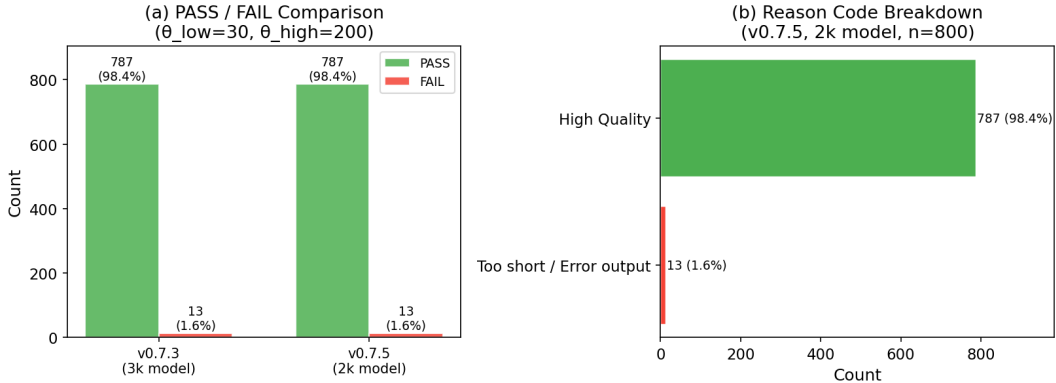


Figure 7: Quality Guard recalibration results (v0.7.3 vs. v0.7.5, $n = 800$). Left: PASS/FAIL verdict comparison (both 98.4% PASS); Right: reason code breakdown for v0.7.5 (2k model).

S.7.1 Structural Member and Damage Type: pairedata v1 vs. v2

To clarify the behavioral change introduced by pairedata v2 (v0.7.3), we replicate the structural member and damage-type analysis of Figure 10 in the main body and apply it to both pipeline versions (Figure 8).

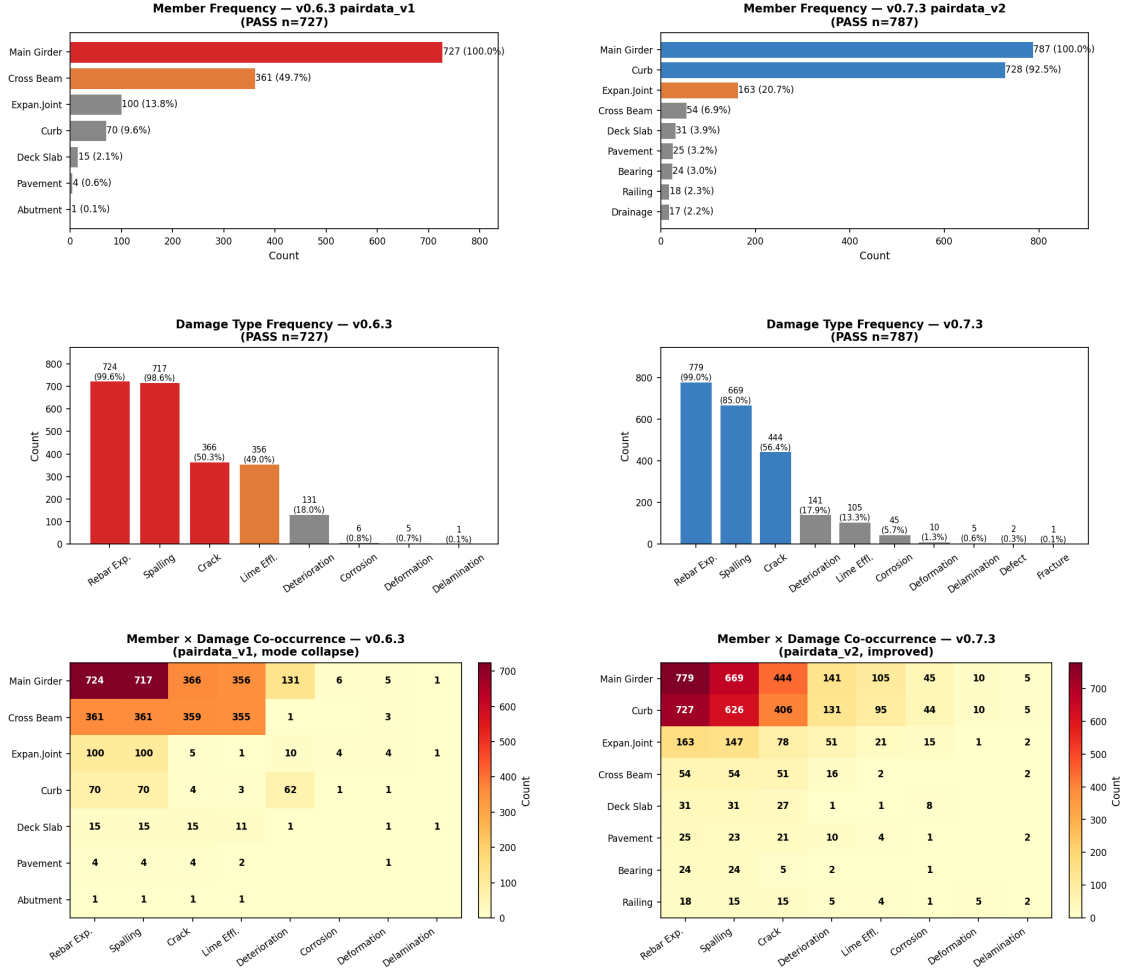


Figure 8: Structural member and damage type analysis for PASS samples: *left column* v0.6.3 pairedata_v1 ($n = 727$), *right column* v0.7.3 pairedata_v2 ($n = 787$). *Top*: member mention frequency; *Middle*: damage type frequency; *Bottom*: member $\times$ damage co-occurrence heatmap.

Member distribution. Both versions share Main Girder dominance (100%), but the secondary-member pattern differs markedly. In v0.6.3, Cross Beam appears as secondary at 49.7%, reflecting a generalized bridge-body description. In v0.7.3, Curb rises to 92.5%, indicating that the 3k model trained on pairedata_v2 now explicitly localizes rebar-exposure damage to the overhang and parapet region. This change is expected: pairedata_v2 was curated to include diverse rebar-exposure examples captured at curb and fascia locations, prompting the model to identify that structural zone.

Damage type distribution. Rebar Exposure remains the dominant keyword in both versions (v0.6.3: 99.6%; v0.7.3: 99.0%). However, the co-occurrence heatmap reveals a structural shift: v0.6.3 assigns nearly identical counts to Main Girder $\times$ Rebar Exposure (724) and Main Girder $\times$ Spalling (717), reflecting the mode-collapsed vocabulary that caused scoring saturation. In v0.7.3, Crack mentions increase to 56.4% (vs. 50.3%), and Lime Efflorescence drops to 13.3% (vs. 49.0%), suggesting a more focused and contextually appropriate description of rebar-exposure-related deterioration.

Discussion: bias suppression through noise removal. The distributional shifts observed above are a direct consequence of the noise-removal strategy applied during pairedata v2 construction. In pairedata v1, the fine-tuning corpus contained out-of-scope descriptions: generic bridge overviews, administrative metadata transcribed from inspection templates, and multi-member narratives that enumerated components regardless of whether they appeared in the target image. The model trained on this data acquired a stereotyped co-occurrence pattern—routinely pairing Main Girder with Cross Beam, Rebar Exposure, and Spalling as a fixed vocabulary template, independent of actual image content. This is evidenced by the nearly equal counts of Main Girder $\times$ Rebar Exposure (724) and Main Girder $\times$ Spalling (717) in v0.6.3, and by the high Lime Efflorescence rate (49.0%) despite the test set consisting primarily of rebar-exposure images where lime efflorescence is not the dominant deterioration form.

In pairedata v2, out-of-scope content was removed so that each training example describes only the damage visible within the image frame. The resulting model exhibits reduced output bias: the near-identical Rebar Exposure–Spalling coupling breaks down (Spalling drops from 98.6% to 85.0%), Lime Efflorescence falls to 13.3%, and the secondary member shifts from the stereotyped Cross Beam to Curb—the structural zone where rebar-exposure damage is physically concentrated in the test images. Together, these changes confirm that noise removal in the training corpus is an effective mechanism for suppressing hallucinated co-occurrence patterns and aligning model output with image-specific damage evidence.

Model scale comparison: v0.7.3 (3k) vs. v0.7.5 (2k). To assess whether reducing the training duration from 3k to 2k steps affects structural member and damage-type vocabulary, we apply the same analysis to v0.7.5 (Figure 9). Main Girder dominance (100%) and Curb co-occurrence (94.7%) are preserved across both scales, confirming that the spatial localization improvement introduced by pairedata v2 is robust to model size. However, a notable divergence appears in the Cross Beam mention rate: the 2k model (v0.7.5) includes Cross Beam in 98.3% of predictions, compared to only 6.9% in the 3k model (v0.7.3). This elevated cross-beam co-occurrence—recalling the pattern observed in v0.6.3 (49.7%)—suggests that shorter training on pairedata v2 retains a partial template pairing Main Girder with Cross Beam as a structural header, even when the target damage is localized to the curb zone. Despite this vocabulary difference, the co-occurrence heatmaps show no systematic shift in damage-member coupling between the two scales, and overall priority scoring behavior (see S.7.2) remains consistent, confirming that pairedata v2 quality is the primary driver of scoring improvement rather than the number of training steps.

Structural Member and Damage Type Analysis: v0.7.3 (3k model) vs v0.7.5 (2k model)

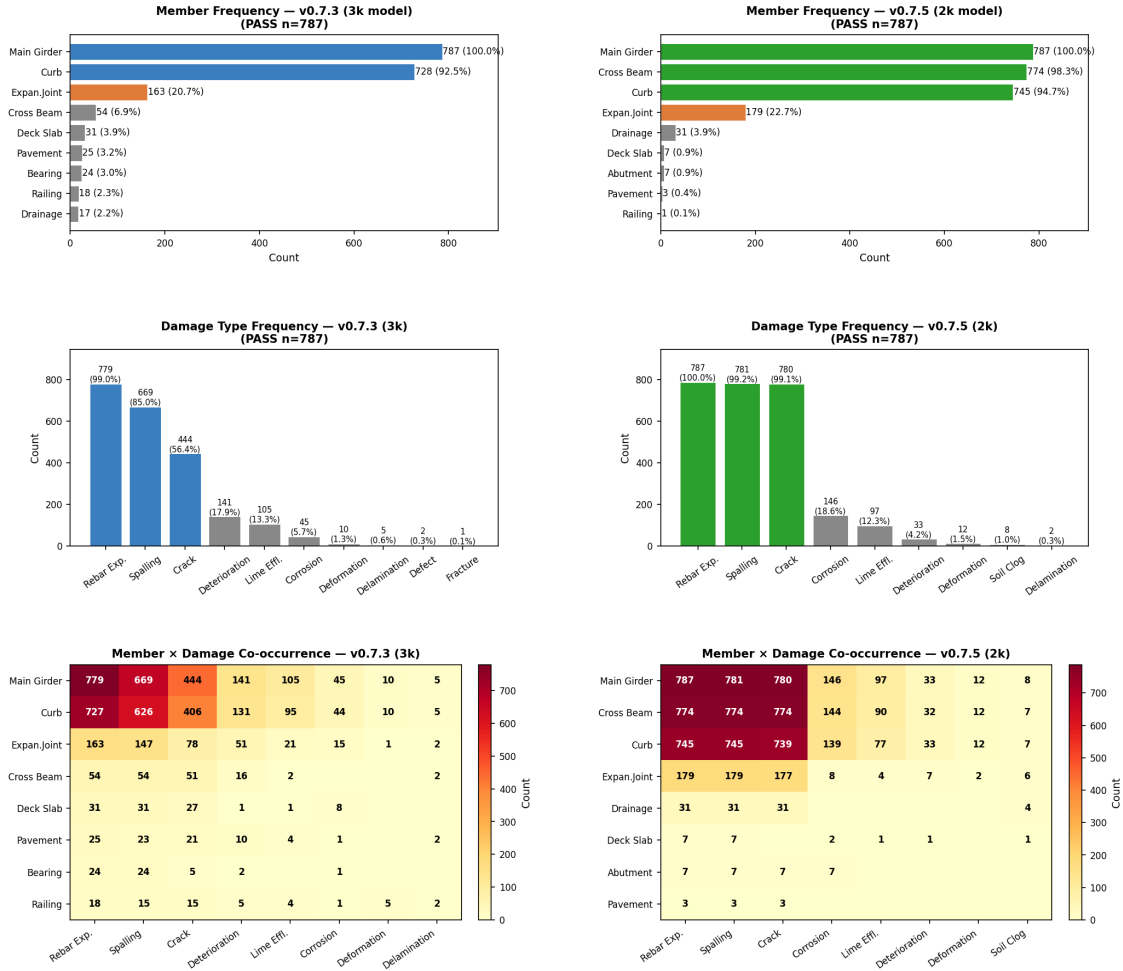


Figure 9: Structural member and damage type analysis: v0.7.3 (3k model) vs. v0.7.5 (2k model) (PASS samples, $n = 787$ each). *Left column:* v0.7.3 (3k); *Right column:* v0.7.5 (2k). *Top:* member mention frequency; *Middle:* damage type frequency; *Bottom:* member $\times$ damage co-occurrence heatmap. Both scales maintain Curb localization ($\geq 92.5\%$); the 2k model additionally shows elevated Cross Beam co-occurrence (98.3 % vs. 6.9 %), suggesting partial retention of a generalized description template.

S.7.2 Priority Score Distribution

Figure 10 presents the continuous priority-score distributions for three pipeline versions using violin plots and stacked level-composition bars.

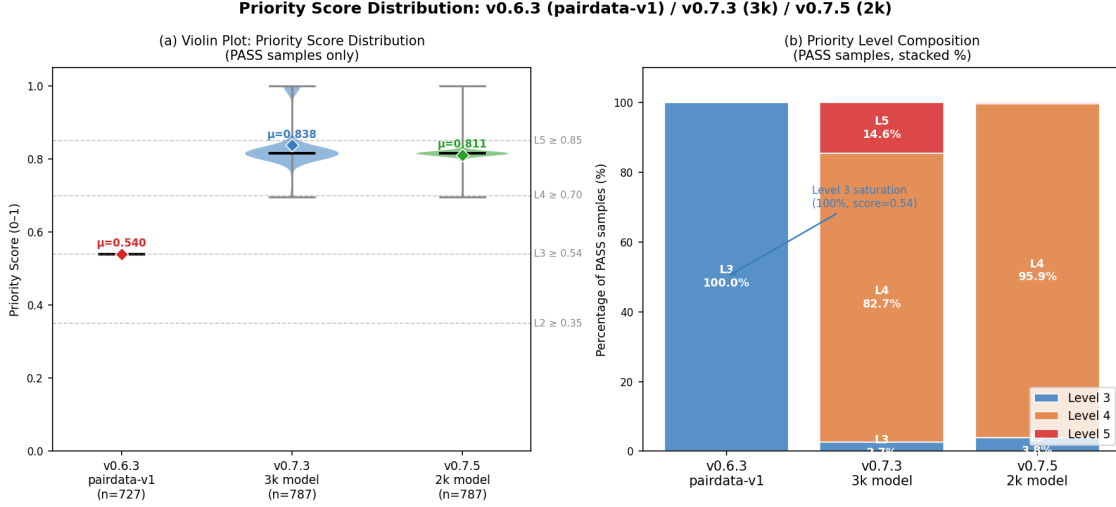


Figure 10: Priority score distribution across three pipeline versions (PASS samples only). *Left (a)*: violin plots with median (horizontal bar) and mean (diamond marker); dashed lines mark priority-level thresholds. *Right (b)*: stacked percentage bars showing Level 3/4/5 composition. v0.6.3 collapses to a point mass at score 0.54 (Level 3, zero variance); both pairedata-v2 models restore score discriminability, concentrating mass in the Level 4 region ($\mu = 0.838$ and $\mu = 0.811$ for v0.7.3 and v0.7.5 respectively).

The violin plot reveals a qualitative shift in the scoring distribution across pipeline versions. Under v0.6.3 (pairedata-v1), the priority score collapses to a point mass at 0.54, producing a degenerate distribution with zero variance. This confirms that the YAML rule engine, operating on mode-collapsed model outputs, cannot distinguish severity levels within the PASS subset.

Both pairedata-v2 models exhibit broad, asymmetric distributions spanning the Level 4 region (0.70–0.85). The v0.7.3 3k model achieves mean score $\mu = 0.838$ (std 0.070), with a longer right tail extending into Level 5 (≥ 0.85), accounting for its 14.6% Level 5 allocation. The v0.7.5 2k model shows a narrower, more concentrated distribution ($\mu = 0.811$, std 0.025), with scores tightly clustered around the median (0.815), reflecting the 2k model’s tendency to produce medium-high severity vocabulary without the extreme-severity descriptors that push scores above the Level 5 threshold.

This comparison demonstrates that both pairedata-v2 models successfully restore score discriminability—from a point-mass degenerate distribution to a Level 4-dominated spread—while exhibiting distinct variance profiles driven by training duration. The 2k model favors operational predictability (consistent Level 4 assignments, std 0.025), whereas the 3k model achieves higher severity coverage at the cost of greater variability (std 0.070).

S.8 Qualitative Analysis: Representative PASS Predictions

Figures 11–13 present twelve representative PASS predictions for each of the two pipeline variants. **Pred₁** is produced by the v0.7.5 pipeline (2k model); **Pred₂** is produced by the v0.7.3 pipeline (3k model), shown for comparison. Both pipelines use Quality Guard $\theta_{\text{low}} = 30$, $\theta_{\text{high}} = 200$. Images are selected to cover a range of structural members and damage types. $\text{Sim}(\text{emb})$ denotes the sentence-embedding cosine similarity for the 2k model output. Shared damage/member terms between GT and **Pred₁** (2k) are bold in **Pred₁**; terms shared between GT and **Pred₂** (3k) are bold in **Pred₂**.



Quality Metrics (2k): $\text{Sim}(\text{emb})$: 0.552

GT: main girder cracking(max W=0.6mm), cross beam cracking(max W=0.25mm), deck slab cracking(W=0.2mm); lime efflorescence observed. cracking , drying shrinkage ; ASR .

Pred₁ (2k): main girder cracking(max W=0.3mm); rebar exposure, cross beam delamination; rebar exposure observed. Also, filler zone lime efflorescence observed. parapet drying shrinkage water leakage, localized. periodic monitoring;; repair if necessary. rebar exposure, insufficient cover and carbonation.

Pred₂ (3k): main girder rebar exposure observed. Also, filler zone , lime efflorescence is present. localized , joint .



Quality Metrics (2k): $\text{Sim}(\text{emb})$: 0.418

GT: RCT girder, cross beam and deck slab delamination, rebar exposure . joint main girder delamination . RCT girder and deck slab spalling corrosion .

Pred₁ (2k): main girder cracking(max W=0.3mm); rebar exposure, cross beam delamination; rebar exposure observed. localized. Also, parapet runoff water insufficient cover carbonation. joint and expansion joint.

Pred₂ (3k): main girder rebar exposure observed. Also, parapet/overhang delamination water leakage observed. parapet , drying shrinkage insufficient cover . localized .



Quality Metrics (2k): $\text{Sim}(\text{emb})$: 0.658

GT: main girder delamination, corrosion, water leakage trace observed. repair as needed.

Pred₁ (2k): main girder cracking(max W=0.3mm); rebar exposure, cross beam delamination; rebar exposure observed. localized. Also, parapet runoff water insufficient cover carbonation. periodic monitoring;; repair if necessary.

Pred₂ (3k): main girder rebar exposure observed. Also, parapet/overhang delamination rebar exposure observed. periodic monitoring;; repair if necessary .



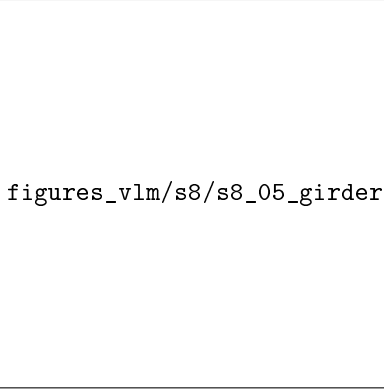
Quality Metrics (2k): $\text{Sim}(\text{emb})$: 0.441

GT: ; main girder expansion joint water leakage due to section thinning corrosion observed. Also, splice bolt corrosion observed. ; cross beam age-related deterioration due to corrosion due to observed.

Pred₁ (2k): main girder cracking(max W=0.3mm); rebar exposure, cross beam delamination; rebar exposure observed. localized. Also, parapet runoff water corrosion.

Pred₂ (3k): main girder rebar exposure observed. Also, parapet/overhang delamination rebar exposure observed. periodic monitoring;; repair if necessary .

Figure 11: Representative PASS predictions (1/3). Each panel: **GT** (original record), **Pred₁** (v0.7.5 2k model), **Pred₂** (v0.7.3 3k model, comparison). $\text{Sim}(\text{emb})$ is for the 2k model. GT/**Pred₁** shared terms bold in **Pred₁**; GT/**Pred₂** shared terms bold in **Pred₂**.



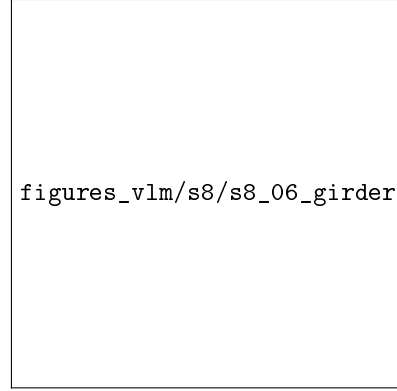
figures_vlm/s8/s8_05_girder_leakage.jpg

Quality Metrics (2k): *Sim(emb): 0.537*

GT: **main girder cracking** drying shrinkage due to age-related deterioration . **main girder** water leakage main girder joint age-related deterioration .

Pred₁ (2k): **main girder cracking**(max W=0.25mm); rebar exposure, cross beam delamination; rebar exposure observed. localized. Also, parapet.

Pred₂ (3k): **main girder** rebar exposure observed. Also, parapet , (localized) .



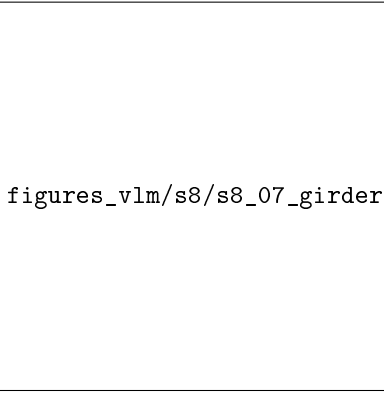
figures_vlm/s8/s8_06_girder_settlement.jpg

Quality Metrics (2k): *Sim(emb): 0.643*

GT: **expansion joint** deformation, P3 ; **main girder** , drainage basin sediment clogging observed. drainage basin , , periodic monitoring;; **repair** if necessary .

Pred₁ (2k): **main girder** cracking(max W=0.3mm); rebar exposure, cross beam delamination; rebar exposure observed. localized. Also, parapet observed, **expansion joint. repair** as needed.

Pred₂ (3k): **main girder** cracking, delamination observed. Also, filler zone rust staining , progression monitoring required . parapet and guardrail rebar exposure , localized rebar exposure , periodic monitoring;; repair if necessary .



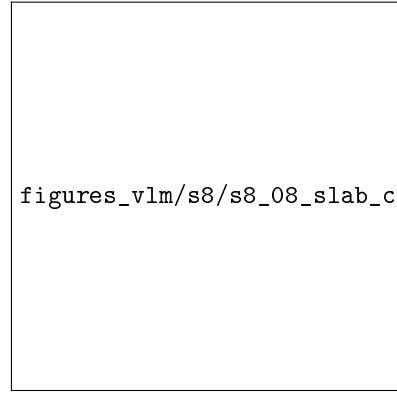
figures_vlm/s8/s8_07_girder_other.jpg

Quality Metrics (2k): *Sim(emb): 0.593*

GT: **main girder cracking, delamination** observed. **repair** is recommended.

Pred₁ (2k): **main girder cracking**(max W=0.3mm); rebar exposure, cross beam **delamination**; rebar exposure observed. localized, periodic monitoring;; **repair** if necessary. Also, expansion joint water leakage.

Pred₂ (3k): **main girder** rebar exposure, **delamination** observed. Also, parapet/overhang observed. periodic monitoring;; repair if necessary . , expansion joint water leakage .



figures_vlm/s8/s8_08_slab_cracking.jpg

Quality Metrics (2k): *Sim(emb): 0.700*

GT: PC slab bridge rubber bearing **cracking**, deformation was observed.

Pred₁ (2k): **main girder cracking**(max W=0.3mm); rebar exposure, cross beam delamination; rebar exposure observed. localized. Also, parapet runoff water insufficient cover carbonation. periodic monitoring;; repair if necessary.

Pred₂ (3k): **main girder** rebar exposure observed. Also, generally water leakage; lime efflorescence , parapet runoff water drying shrinkage due to **cracking** . rebar exposure insufficient cover carbonation .

Figure 12: Representative PASS predictions (2/3). Each panel: **GT** (original record), **Pred₁** (v0.7.5 2k model), **Pred₂** (v0.7.3 3k model, comparison). *Sim(emb)* is for the 2k model. GT/**Pred₁** shared terms bold in **Pred₁**; GT/**Pred₂** shared terms bold in **Pred₂**.

figures_vlm/s8/s8_09_slab_rebar_exposure.jpg



Quality Metrics (2k): *Sim(emb): 0.555*

GT: deck slab **cracking** was observed. Also, due to **rebar exposure** was observed, monitoring is required.

Pred₁ (2k): main girder **cracking**(max W=0.3mm); **rebar exposure**, cross beam delamination; **rebar exposure** observed, localized. Also, parapet runoff water insufficient cover carbonation. periodic monitoring;; repair if necessary.

Pred₂ (3k): main girder **cracking**, delamination, **rebar exposure** observed. **deck slab** localized rebar exposure, expansion joint parapet runoff water.

Quality Metrics (2k): *Sim(emb): 0.770*

GT: A1abutment **cracking**(max W=1.2mm); lime efflorescence, P1pier **cracking**(max W=0.3mm); rust staining lime efflorescence;; A1abutment; P1pier **expansion joint water leakage** trace observed. **cracking** rust staining and localized, **periodic monitoring**;; **repair** if necessary.

Pred₁ (2k): main girder **cracking**(max W=0.3mm); rebar exposure, cross beam delamination; rebar exposure observed. Also, parapet **expansion joint water leakage** due to insufficient cover observed. age-related deterioration drying shrinkage. **periodic monitoring**;; **repair** if necessary.

Pred₂ (3k): main girder rebar exposure observed. Also, parapet/overhang delamination **water leakage** is present. parapet delamination, insufficient cover carbonation. localized.



Quality Metrics (2k): *Sim(emb): 0.798*

GT: abutment, pier **cracking**, **rebar exposure**, water leakage trace observed. from a durability standpoint, **repair** is recommended.

Pred₁ (2k): main girder **cracking**(max W=0.3mm); **rebar exposure**, cross beam delamination; **rebar exposure** observed. Also, parapet expansion joint infiltration water due to rust staining. localized, periodic monitoring;; **repair** if necessary.

Pred₂ (3k): main girder **rebar exposure** observed. Also, generally **cracking** delamination, expansion joint infiltration water internal rebar expansion.



Quality Metrics (2k): *Sim(emb): 0.656*

GT: abutment honeycombing spalling is present. **bearing** soil ejection, lime efflorescence water leakage trace observed. abutment honeycombing, rust staining water leakage trace observed. observed.

Pred₁ (2k): main girder **cracking**(max W=0.3mm); rebar exposure, cross beam delamination; rebar exposure observed. localized, periodic monitoring;; repair if necessary. Also, parapet runoff water corrosion.

Pred₂ (3k): main girder rebar exposure, delamination observed. Also, parapet/overhang, expansion joint infiltration water **bearing**. periodic monitoring;; repair if necessary. , expansion joint infiltration water due to.

Figure 13: Representative PASS predictions (3/3). Each panel: **GT** (original record), **Pred₁** (v0.7.5 2k model), **Pred₂** (v0.7.3 3k model, comparison). *Sim(emb)* is for the 2k model. GT/Pred₁ shared terms bold in Pred₁; GT/Pred₂ shared terms bold in Pred₂.

When damage/member keywords appear in *both* GT and Pred₂ (3k), *Sim(emb)* is correspondingly high, indicating successful damage recognition. Pred₁ (2k) consistently includes cross beam (98.3 % rate) and the cracking-rebar-exposure compound descriptor, reflecting the 2k model's template-driven vocabulary. Conversely, terms present only in GT are absent from both predictions—representing the remaining recognition gap and a direction for future improvement.